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Canteen Management System

Subject Name: Web Development
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INTRODUCTION

The Canteen Management System is a web-based application designed to automate and streamline canteen operations within educational institutions. It allows students and staff to view menus, place orders, and make payments digitally, while administrators can efficiently manage menus, track orders, and handle billing through a centralized platform.

The system reduces manual effort, eliminates human errors, and maintains accurate real-time data on food items, prices, and stock availability. By integrating both frontend and backend technologies, the project ensures faster service, better communication, and smooth management of daily canteen activities. Overall, it aims to create a more convenient, transparent, and efficient canteen experience for everyone.

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PROBLEM STATEMENT

In Universities, Canteen operations are managed manually, which leads to various challenges such as:

- Delay in taking and delivering orders
- Mismanagement of inventory and stock
- Lack of proper communication between staff and customers
- Time-consuming manual billing and order tracking

To overcome these issues, a computerized full-stack Canteen Management System with backend and database support was developed to automate ordering, billing, and stock management efficiently.

BACKGROUND STUDY

In educational institutions and workplaces, canteens play an essential role in providing food and refreshments to students and staff. Traditionally, these canteens rely on manual processes for recording orders, updating menus, managing inventory, and calculating bills.

However, as the number of customers increases, these manual methods become inefficient, error-prone, and time-consuming. Managing large volumes of orders, keeping track of stock, and ensuring timely service becomes difficult without a systematic approach.

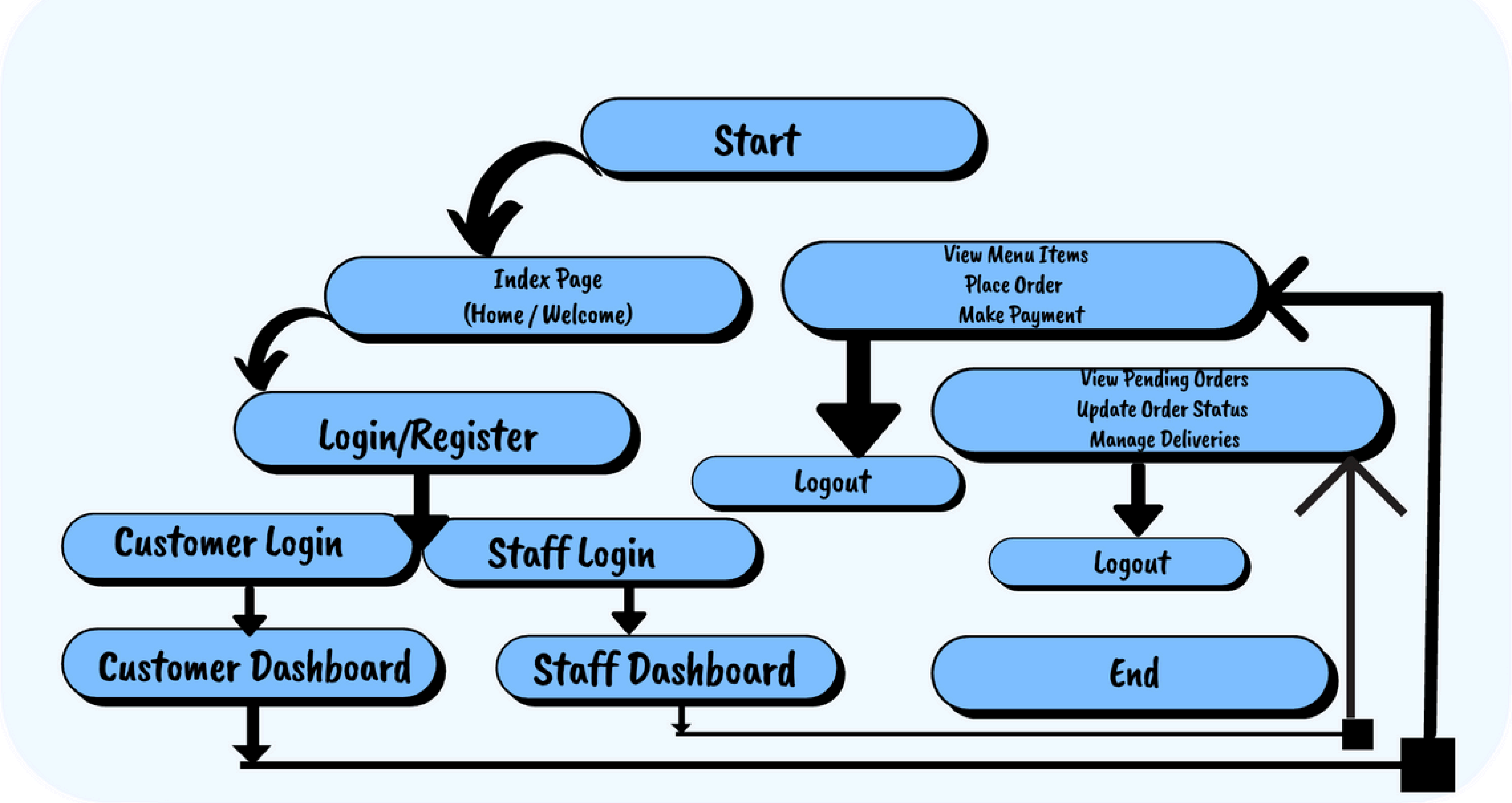
With the advancement of technology and the growing importance of digital transformation, automating canteen operations through a web-based system has become necessary. A digital platform can streamline the entire workflow — from order placement to billing — making the process faster and more accurate.

This background led to the development of the Canteen Management System, a web application designed to modernize canteen management and enhance user experience for both customers and staff.

FEATURES

- **User Authentication (Login & Registration)** with database validation
- **Dynamic Menu Management** — items stored in backend database
- **Order Management** with real-time updates
- **Backend API** Integration for fetching and storing data
- **Persistent Data Storage** — all menu items, orders, and users saved in database (JSON/server/DB)
- **Admin Dashboard** for managing orders and stock

FLOW CHART

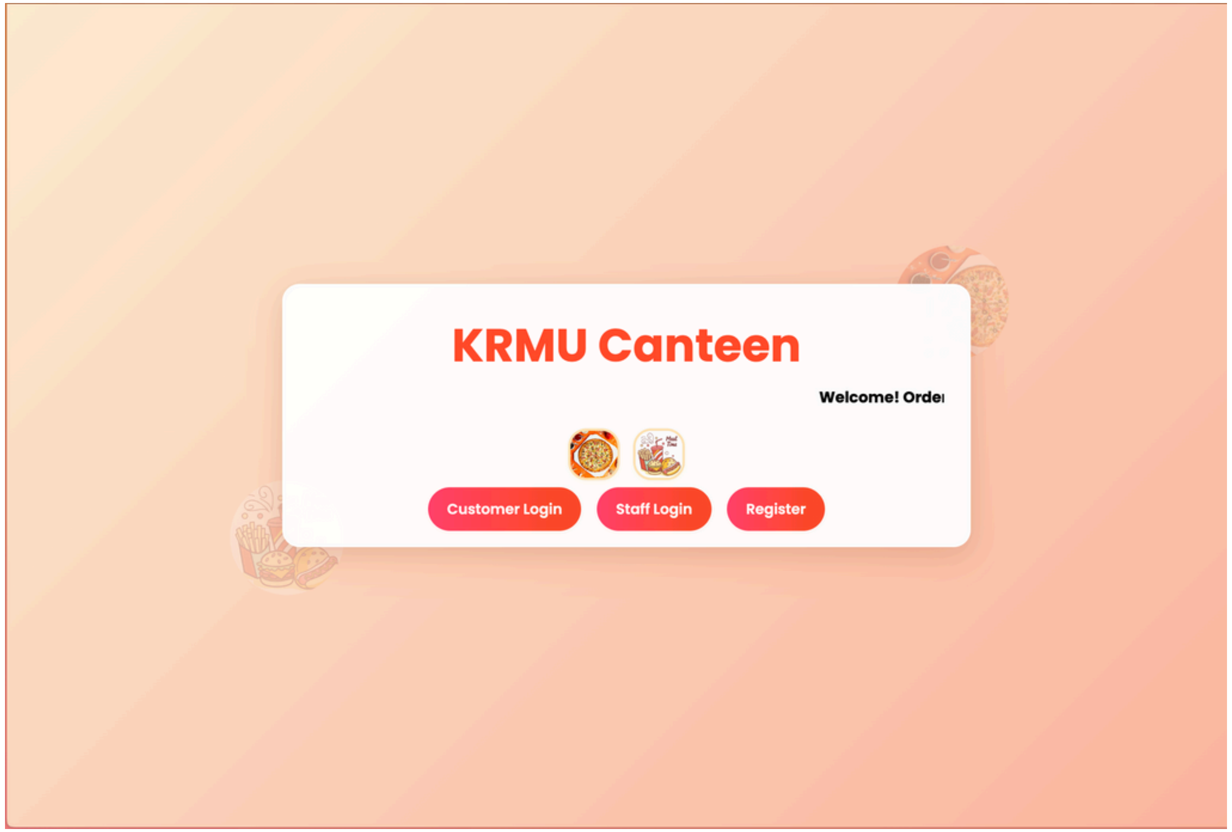


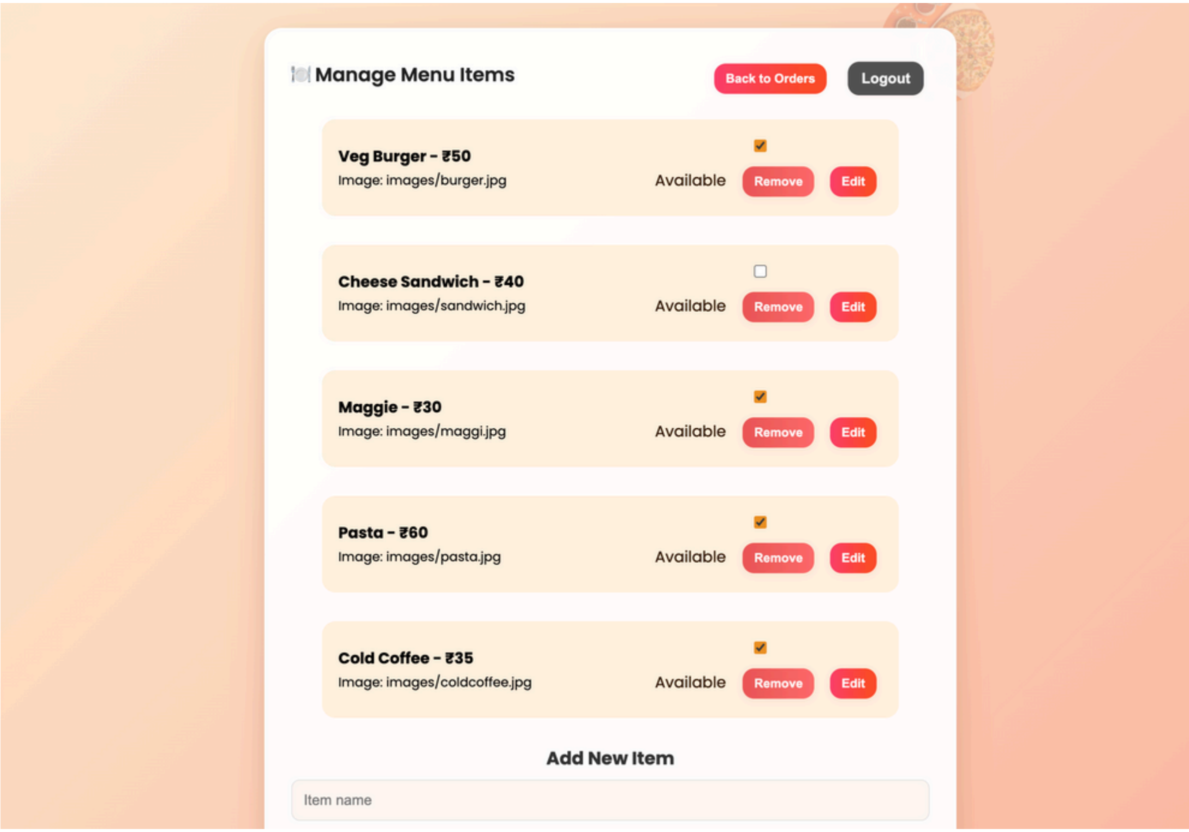
Start
Index Page
(Home / Welcome)
Login/Register
Customer Login
Staff Login
Customer Dashboard
Staff Dashboard
View Menu Items
Place Order
Make Payment
Logout
View Pending Orders
Update Order Status
Manage Deliveries
Logout
End

TECHNOLOGIES USED

- **Frontend Technologies**
  - HTML – For creating the structure of web pages
  - CSS – For styling and responsive design
  - JavaScript – For adding interactivity and dynamic functionality
- **Backend (Logic Layer)**
  - Node.js and Express.js (Server-side logic & API handling)
- **Database / Storage**
  - JSON-based storage / MongoDB / Local file database
- **Development Tools**
  - Visual Studio Code – For writing and debugging code
  - Git & GitHub – For version control and collaboration
  - Netlify – For hosting and deploying the web application
  - Render / Localhost (Backend hosting)

OUTPUT


KRMU Canteen
Welcome! Orde
Customer Login
Staff Login
Register


Manage Menu Items
Back to Orders
Logout
Veg Burger - ₹50
Image: images/burger.jpg
Available
Remove
Edit
Cheese Sandwich - ₹40
Image: images/sandwich.jpg
Available
Remove
Edit
Maggie - ₹30
Image: images/maggi.jpg
Available
Remove
Edit
Pasta - ₹60
Image: images/pasta.jpg
Available
Remove
Edit
Cold Coffee - ₹35
Image: images/coldcoffee.jpg
Available
Remove
Edit
Add New Item
Item name

FUTURE SCOPE

- **Online Payment Integration**
  Add secure payment gateways like UPI, Paytm, or credit/debit cards for seamless transactions.
- **AI-Based Recommendations**
  Use Artificial Intelligence to suggest popular or frequently ordered items to customers.
- **Real-Time Order Notifications**
  Notify users instantly when their order is prepared or ready for pickup.
- **Feedback & Rating System**
  Allow users to give feedback on food and service to help improve quality.
- **Integration**
  With a real-time database (like Firebase or MongoDB Atlas)
- **Cloud deployment**
  backend for 24×7 accessibility

RESULT

The Canteen Management System was successfully upgraded into a full-stack web application with backend functionality. The new system stores and retrieves data dynamically from the backend, ensuring better scalability and reliability."

The system provides a user-friendly interface for customers to place orders and for administrators to manage the entire canteen process digitally. It also generates accurate sales reports and maintains real-time records of available stock.

Overall, the project achieved its objective of reducing manual work, saving time, and improving accuracy and efficiency in canteen management.

LEARNING OUTCOMES

1. **Enhanced Web Development Skills** – Gained hands-on experience in building a full-fledged web application using HTML, CSS, and JavaScript.
2. **Understanding of Frontend–Backend Interaction** – Learned how different components of a web app communicate and manage data efficiently.
3. **Improved Problem-Solving Abilities** – Developed logical thinking to design efficient solutions for real-life problems like digital order management.
4. **Team Collaboration** – Experienced effective teamwork, version control using GitHub, and project coordination.
5. **Project Deployment Knowledge** – Learned how to host and deploy a website on Netlify for public access.
6. **Learned backend development** -Using Node.js and Express.js

CONCLUSION

The Canteen Management System successfully automates and simplifies the daily operations of a canteen. It replaces the traditional manual process with a digital platform that efficiently handles menu updates, order placement, billing, and inventory management.

By using this system, time is saved, errors are minimized, and both staff and customers experience smoother and faster service. It also helps administrators maintain accurate records and generate sales reports for better decision-making.

Overall, this project demonstrates how technology can streamline real-world processes, improving efficiency, transparency, and user satisfaction in canteen operations.

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These references were used for research, understanding technologies, and designing the Canteen Management System project.

THANK YOU

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